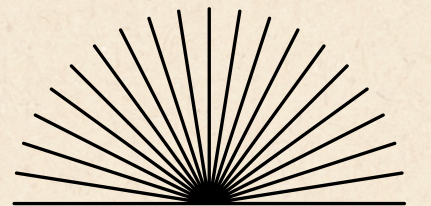


COPERNICUS SENTINEL 2

RELATIONAL DATABASE

Data Management for Data Science a.a. 2025-2026

Alfieri Carlo Alberto (2263545), Sorrentino Mariapia (2268884)



DATA PROFILE

ESA Copernicus Stac API

- Sensor statistics and image urls
- Spatial and temporal information of the acquisitions
- Macro environmental statistics

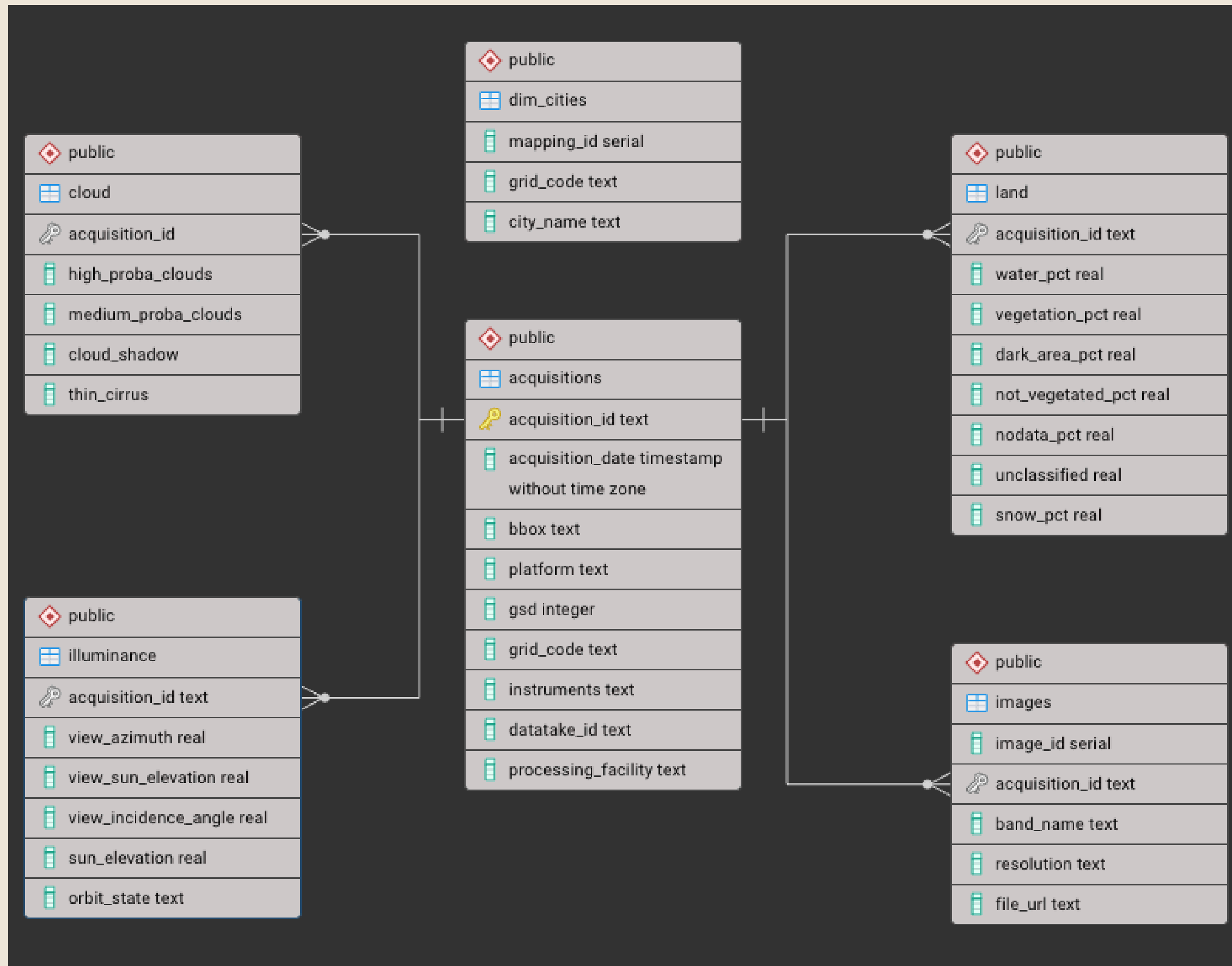
RELATIONAL DATABASE PURPOSE

Historical repository to :

- Retrieve easily good quality images
- Conduct aggregate macro regional analyses
- Check for satellite capabilities and performances



DATABASE SCHEMA



BEFORE & AFTER: OPTIMIZATION RESULTS

QUERY TYPE	PRE OPT RUNTIME	OPTIMIZATION METHOD	POST OPT RUNTIME
Best months per city to acquire satellite images	• avg query time: 184 ms • avg execution time: 1.5 ms • avg planning time: 3.29 ms	Added PK/FK constraints & B-Tree index on FKs	• avg query time: 143.67 ms (-4.30 ms) • avg execution time: 0.32 ms (-1.18 ms) • avg planning time: 0.29 ms (-3 ms)
Seasonality index trends per city	• avg query time: 140.33 ms • avg execution time: 0.11 ms • avg planning time: 0.45 ms	Built Materialized View	• avg query time: 210 ms (+69.67 ms) • avg execution time: 0.03 ms (-0.08 ms) • avg planning time: 0.04 ms (-0.41 ms)
Data loss event	• avg query time: 178.66 ms • avg execution time: 1.26 ms • avg planning time: 22.96 ms	Composite Index with time and platform	• avg query time: 150.67 ms (+84 ms) • avg execution time: 1.81 ms (-0.74 ms) • avg planning time: 0.26 ms (-0.30 ms)
Images extraction via cloud filtering	• -- avg query time: 344 ms • -- avg execution time: 0.61 ms • -- avg planning time: 0.21 ms	Composite Index & LEFT JOIN ... IS NULL Anti-Join	• -- avg query time: 275.66 ms (-68.34 ms) • -- avg execution time: 0.43 ms (-0.18 ms) • -- avg planning time: 0.35 ms (+0.14 ms)

THE END

Thank Your For Listening